



Build a Complete Weather Station with Raspberry Pi 4 Model B

Better than John's weather forecasting stone

In this DIY article, we will demonstrate how to create a weather station that can provide real-time data on temperature, humidity, atmospheric pressure, air quality, wind speed, wind gusts, wind direction, and rainfall using a Raspberry Pi 4 Model B board. We will cover the hardware and software requirements, assembly instructions, circuit

diagrams, programming code, and operation instructions.

Section 1: Hardware

To build your weather station, you will need the following components:

1. Raspberry Pi 4 Model B
2. Weather sensors:
 - DHT22 (Temperature and humidity)

- BMP280 (Atmospheric pressure)
 - SDS011 (Air quality)
 - Anemometer (Wind speed and direction)
 - Rain gauge (Rainfall)
3. Breadboard
 4. Jumper wires (male-to-female and male-to-male)
 5. Resistors
 6. Power supply (5V, 2.5A)
 7. MicroSD card (minimum 16GB, with Raspberry Pi OS)
 8. Enclosure (weatherproof)



Section 2: Software Requirements

To program your weather station, you will need the following software: